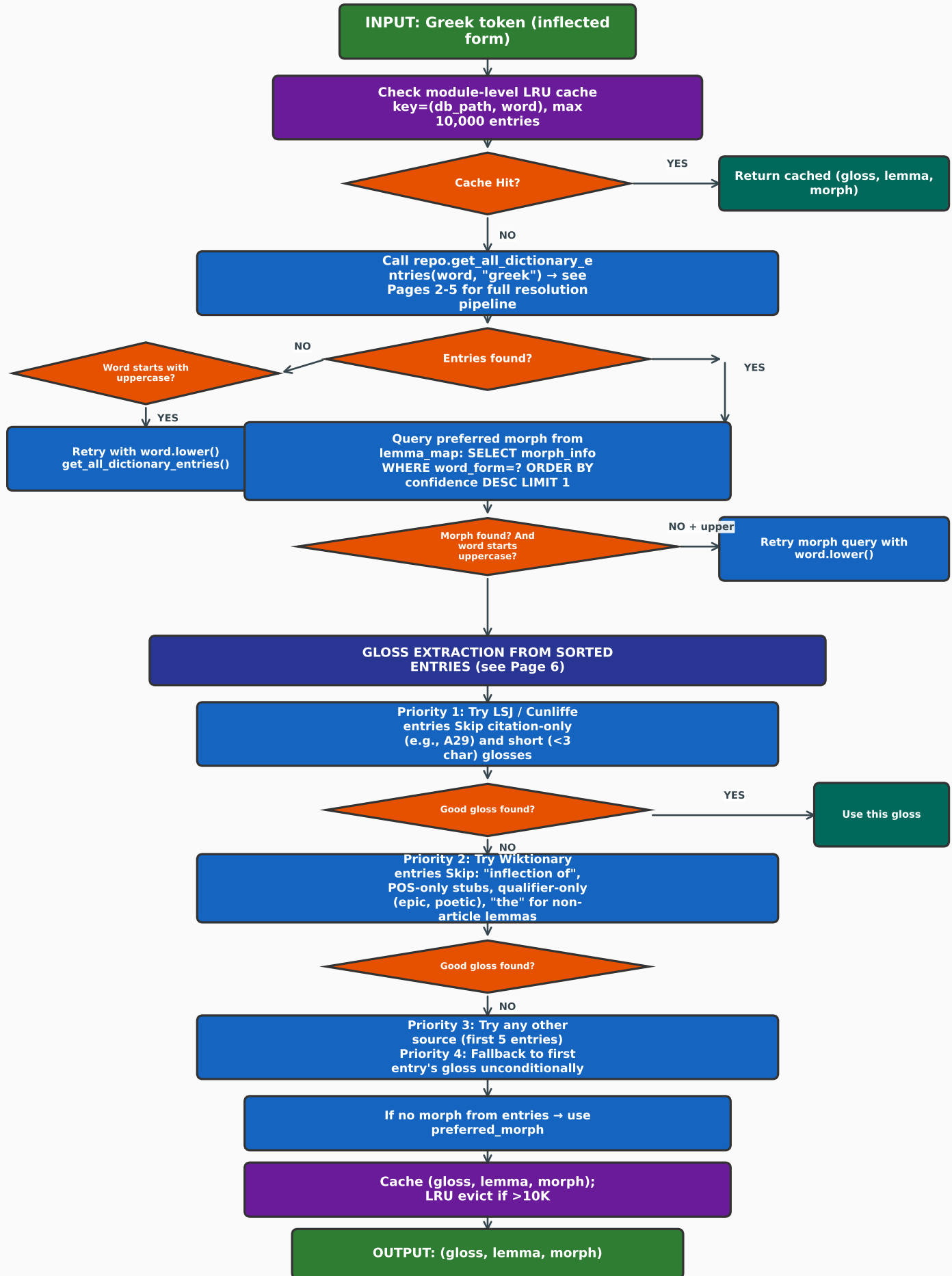


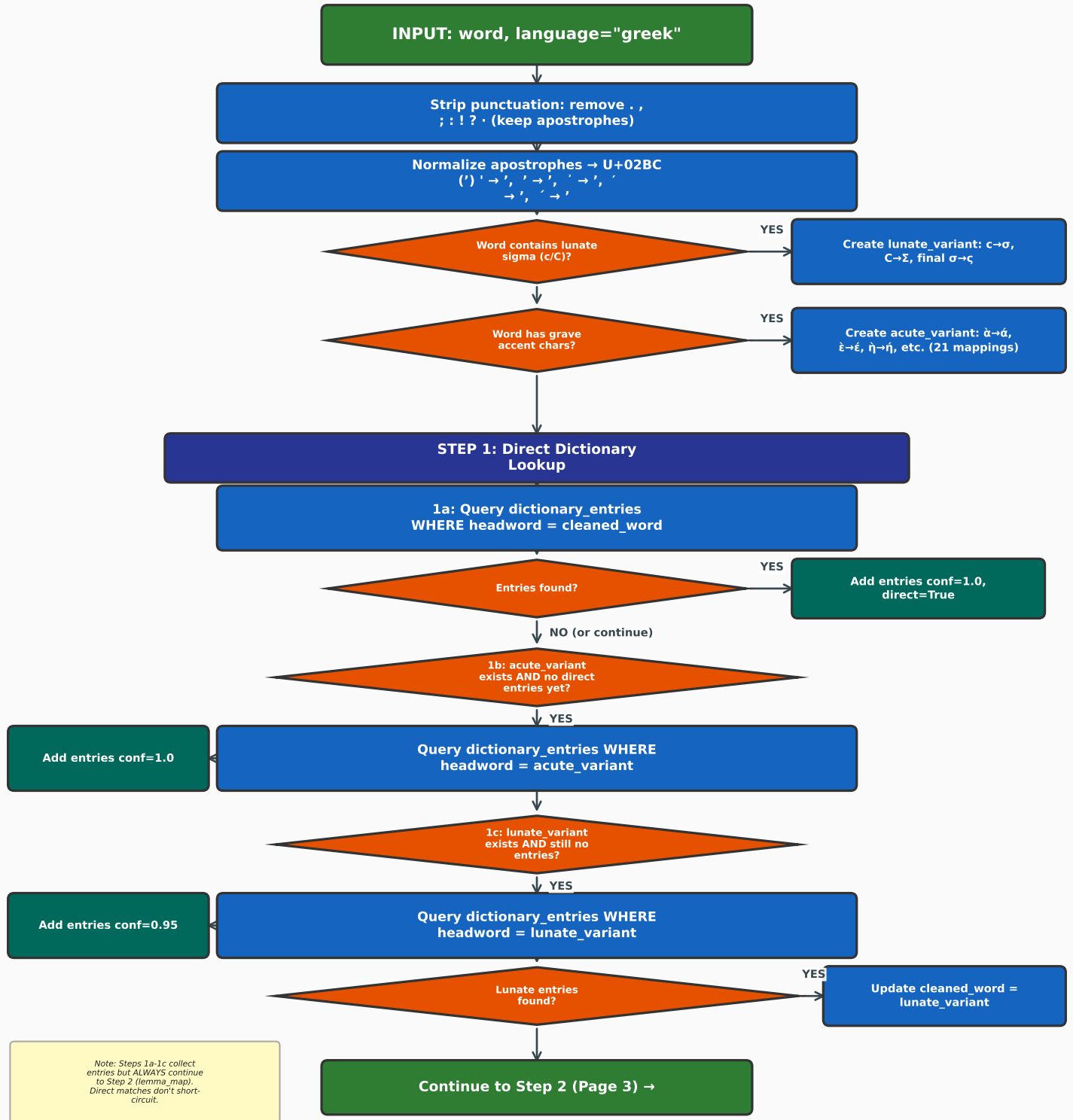
Greek Interlinear Lemma Resolution

Page 1: Entry Point — `_cached_lookup_word()`



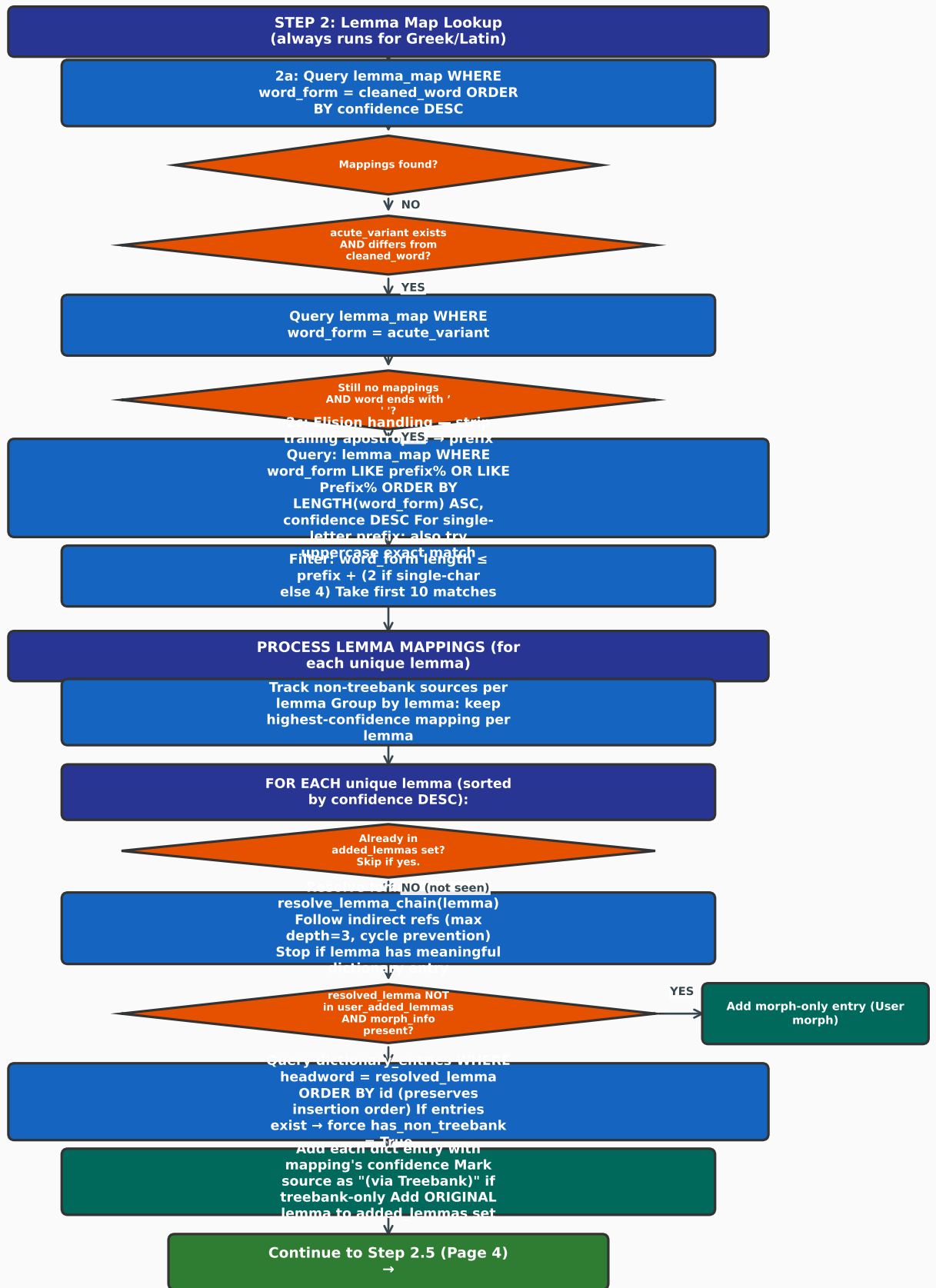
get_all_dictionary_entries() — Preprocessing & Step 1

Page 2: Normalization, Direct Dictionary Lookup



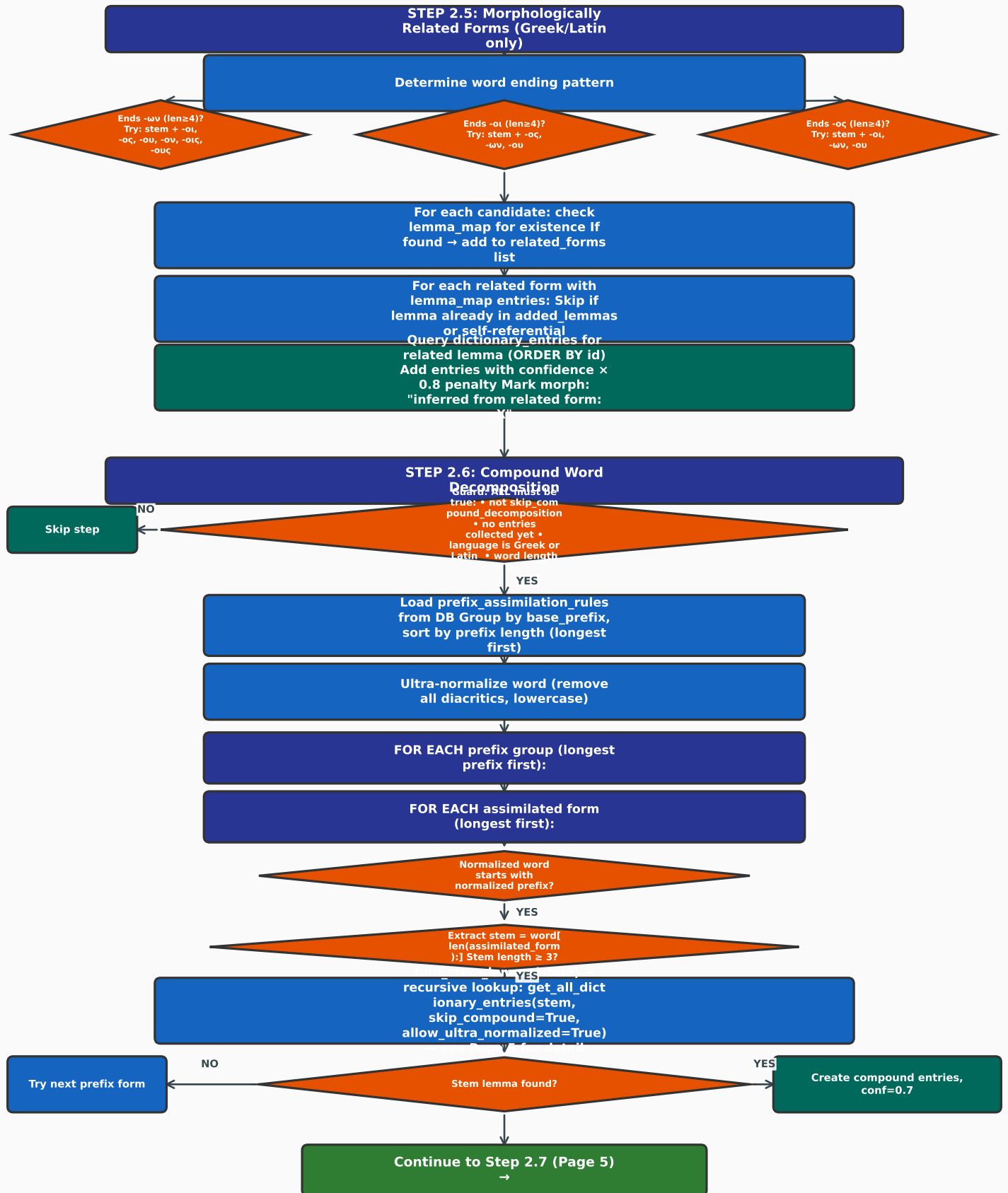
get_all_dictionary_entries() — Step 2

Page 3: Lemma Map Lookup (inflected form → lemma)



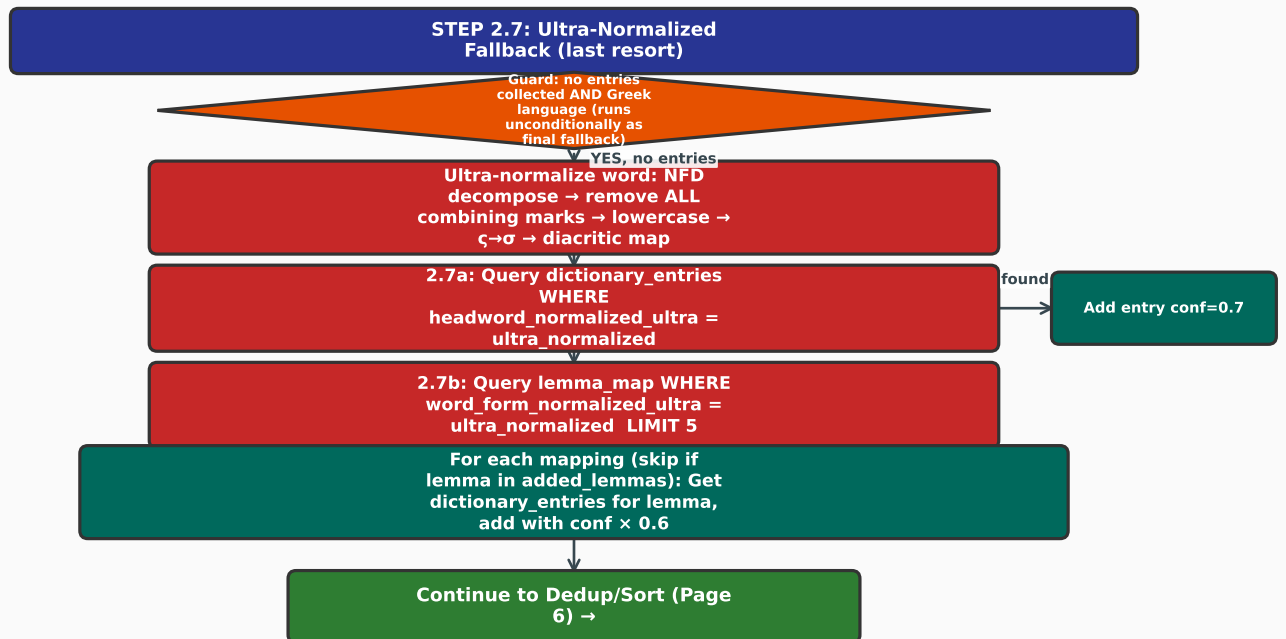
get_all_dictionary_entries() — Steps 2.5 & 2.6

Page 4: Related Forms & Compound Decomposition



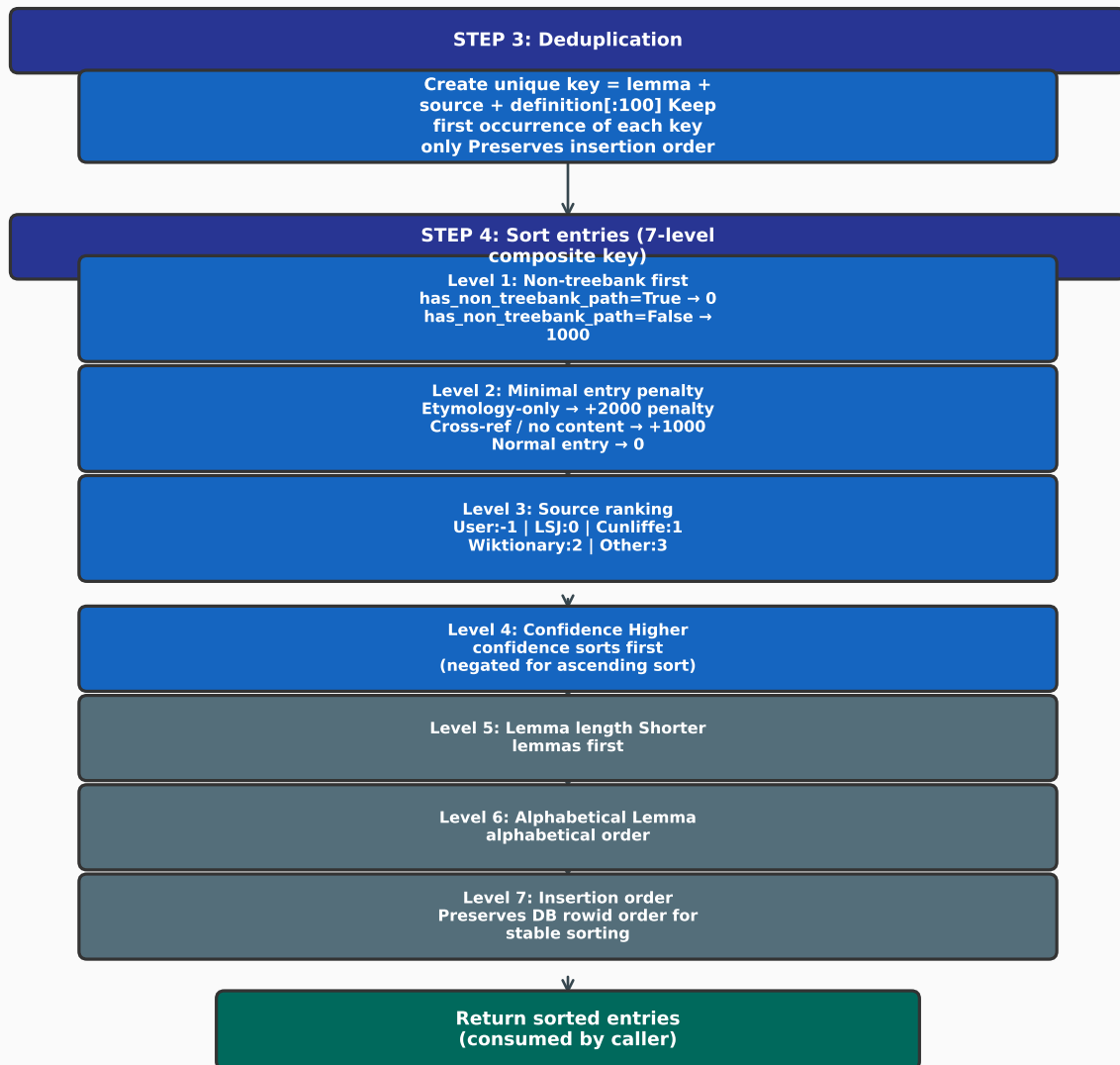
get_all_dictionary_entries() — Steps 2.6 (detail) & 2.7

Page 5: Stem Lemma Resolution & Ultra-Normalized Fallback



get_all_dictionary_entries() — Steps 3 & 4

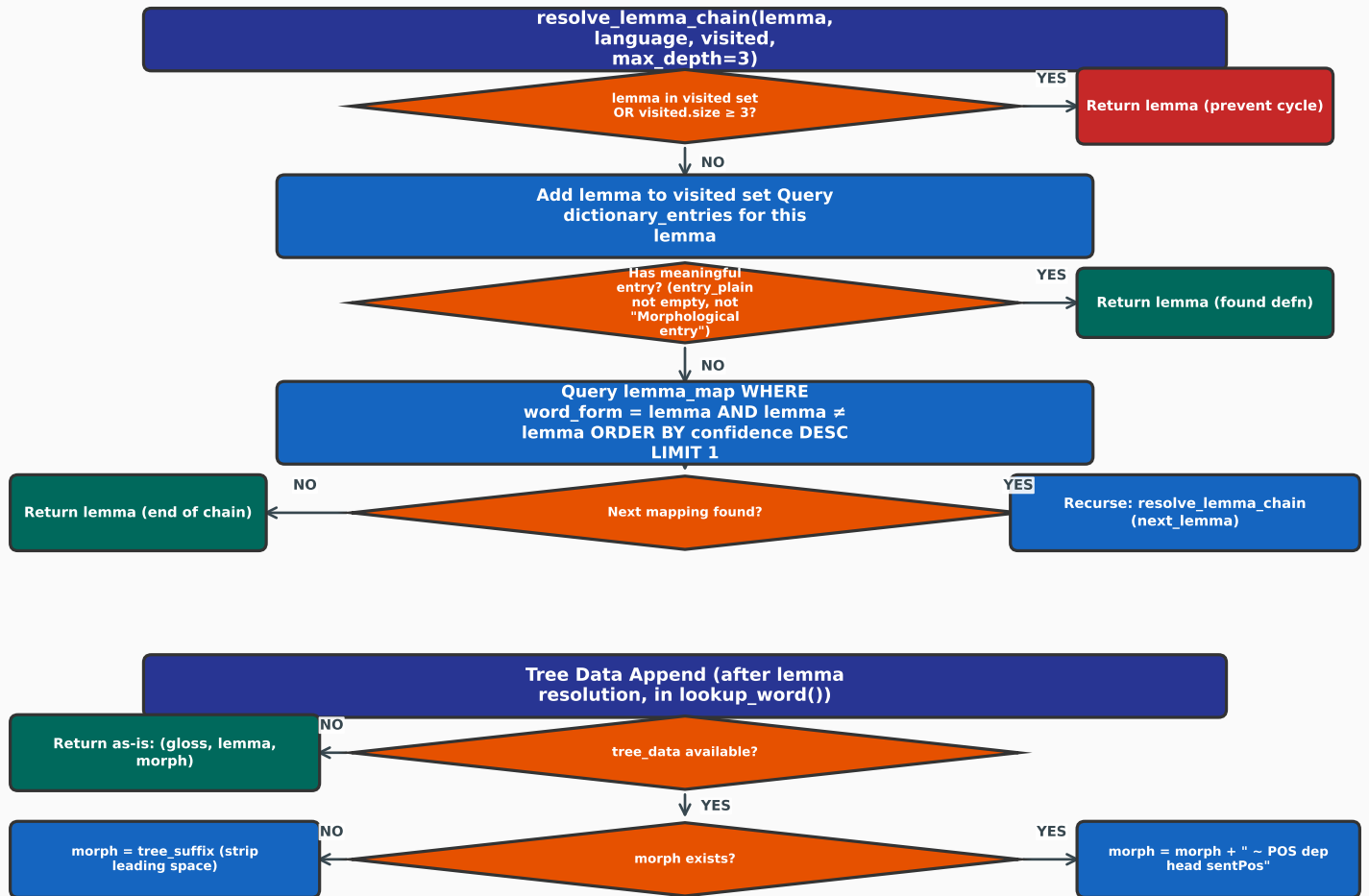
Page 6: Deduplication & Sorting



Level 2 Detail — "Minimal entry" detection: • Strip HTML tags from definition → plain text • Etymology-only: starts with "Etymology:" or "+", no definition sections (A., I., 1.) after • Cross-reference: remaining content has no 3+ letter English words • This ensures real definitions always rank above stubs and cross-references

Lemma Chain Resolution & Complete Summary

Page 7: resolve_lemma_chain() detail, Tree Data, Reference



Complete Resolution Order with Decision Points

Cache → hit? return : miss → proceed
Step 1a: Direct dict lookup (word) → found? add conf=1.0
Step 1b: No direct? acute variant exists? → try dict(acute) → add conf=1.0
Step 1c: Still none? lunate sigma? → try dict(lunate) → add conf=0.95, update word
Step 2a: lemma_map(word) → mappings? → resolve chain → get dict for lemma
Step 2b: No map? acute variant? → try lemma_map(acute)
Step 2c: No map? word ends '?' → strip, prefix search, filter by length, take 10
Step 2.5: Greek endings -ω/-οι/-ος → swap endings → lemma_map → dict, conf×0.8
Step 2.6: No entries, word≥6 → prefix rules → extract stem → recursive lookup
 ↳ find_stem_lemma: try stem → plain vowels → rough breathing vowels → None
Step 2.7: Still none, Greek → ultra-normalize → dict(ultra) conf=0.7 → map(ultra) conf×0.6
Dedup: key=(lemma,source,def[:100])
Sort: non-treebank > minimal-penalty > source-rank > confidence > len > alpha > order

Source: ui_dictionary_lookup.py (PerseusRepository), generate_interlinear.py (_cached_lookup_word)